

Foundations of Chemistry

Energy and Change

Sample Edition for Testing ExamScribe

This short text was written for software testing. It follows the structure of an introductory chemistry textbook.

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Preface

This sample book introduces energy, heat, and enthalpy. Each section begins with learning objectives, defines key terms in bold, and ends with review questions. Answers to the review questions appear in the Answer Key.

Energy and Its Units

Every chemical and physical change involves energy. This chapter introduces the main forms of energy and the units used to measure it.

1.1 What Is Energy?

Learning Objectives

By the end of this section, you will be able to:

- Define energy and distinguish kinetic energy from potential energy
- Calculate the kinetic energy of a moving object
- Describe thermal energy as a form of kinetic energy

Chemical changes and physical changes are almost always accompanied by changes in energy.

Energy is the capacity to supply heat or do work. Work is done when a force moves matter through a distance. Energy exists in many forms, but most of them can be grouped into two broad classes.

Kinetic energy is the energy that an object possesses because of its motion. A rolling ball, a flowing river, and a vibrating molecule all have kinetic energy. The kinetic energy of an object depends on both its mass and its speed:

$$KE = \frac{1}{2}mv^2 \quad (1.1)$$

where m is the mass of the object in kilograms and v is its speed in meters per second. Because the speed is squared, doubling the speed of an object multiplies its kinetic energy by four.

Potential energy is the energy that an object possesses because of its position, composition, or condition. A ball held above the ground has gravitational potential energy, and a battery stores chemical potential energy in the arrangement of its atoms.

Energy can be converted from one form to another, but it is never created or destroyed. This principle is called the **law of conservation of energy**.

The atoms and molecules in any sample of matter are in constant random motion. **Thermal energy** is the kinetic energy associated with the random motion of these particles. A cup of hot tea has more thermal energy than the same cup of tea after it has cooled, because its molecules move faster on average.

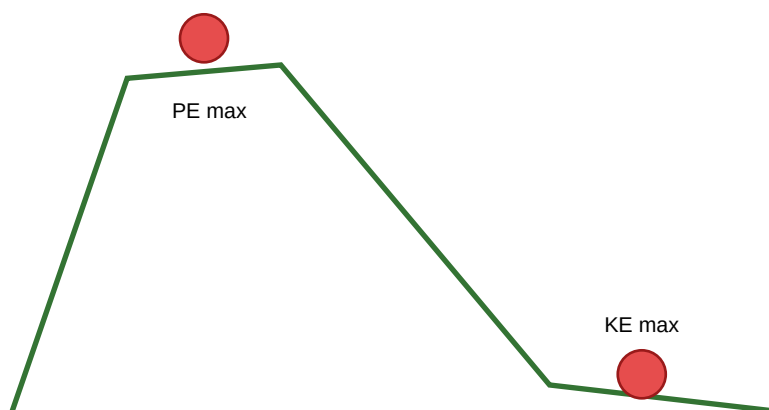


Figure 1.1 A ball at the top of a hill has maximum potential energy. As it rolls down, potential energy is converted to kinetic energy.

Example 1.1 Calculating Kinetic Energy

A 0.500-kg ball rolls at a speed of 4.00 m/s. What is its kinetic energy?

Solution Substitute the mass and speed into Equation 1.1: $KE = \frac{1}{2} \times 0.500 \text{ kg} \times (4.00 \text{ m/s})^2 = 4.00 \text{ J}$.

1.2 Units of Energy

Learning Objectives

By the end of this section, you will be able to:

- Identify the joule as the SI unit of energy
- Convert between joules and calories

The SI unit of energy is the **joule** (J). One joule is the kinetic energy of a 2-kg object moving at a speed of 1 meter per second. A joule is a small amount of energy, so chemists often report energies in kilojoules (kJ); $1 \text{ kJ} = 1000 \text{ J}$.

An older unit, the **calorie** (cal), was originally defined as the amount of energy needed to raise the temperature of 1 gram of water by 1 degree Celsius. Today the calorie is defined in terms of the joule:

$$1 \text{ cal} = 4.184 \text{ J (exactly)} \quad (1.2)$$

The Calorie (with a capital C) that appears on food labels is actually a kilocalorie: 1 Calorie = 1 kcal = 1000 cal.

Example 1.2 Converting Calories to Joules

A snack releases 250 cal of energy when it is burned. How many joules is this?

Solution Multiply by the conversion factor from Equation 1.2: $250 \text{ cal} \times 4.184 \text{ J/cal} = 1046 \text{ J}$, or about 1.05 kJ.

Key Terms

calorie (cal) unit of energy; 1 cal is defined as exactly 4.184 J

energy capacity to supply heat or do work

joule (J) SI unit of energy; $1 \text{ J} = 1 \text{ kg}\cdot\text{m}^2/\text{s}^2$

kinetic energy energy of an object due to its motion

law of conservation of energy energy is converted from one form to another but is never created or destroyed

potential energy energy of an object due to its position, composition, or condition

thermal energy kinetic energy associated with the random motion of atoms and molecules

Key Equations

$$KE = \frac{1}{2}mv^2$$

$$1 \text{ cal} = 4.184 \text{ J}$$

Summary

1.1 What Is Energy?

Energy is the capacity to supply heat or do work. Kinetic energy is energy of motion, and potential energy is energy of position, composition, or condition. Thermal energy is the kinetic energy of randomly moving particles.

1.2 Units of Energy

The joule is the SI unit of energy. One calorie equals exactly 4.184 joules.

Review Questions

1.1 What Is Energy?

1. Which of the following has the greatest kinetic energy? (a) a 1-kg ball at rest (b) a 1-kg ball moving at 2 m/s (c) a 1-kg ball moving at 1 m/s (d) a 2-kg ball at rest
2. Calculate the kinetic energy of a 2.00-kg object moving at 3.00 m/s.

1.2 Units of Energy

3. Convert 500. cal to joules.
4. How many calories are in 1.00 kJ?

Thermochemistry

In Chapter 1 you learned that energy is the capacity to supply heat or do work, and that the joule is the SI unit of energy. This chapter applies those ideas to the heat that flows during chemical and physical changes, and to the thermal energy stored in matter.

2.1 Heat and Temperature

Learning Objectives

By the end of this section, you will be able to:

- Distinguish between heat and temperature
- Identify a system and its surroundings
- Classify processes as endothermic or exothermic

Temperature is a measure of the average kinetic energy of the particles in a sample of matter. **Heat** (q) is the transfer of thermal energy between two bodies at different temperatures. Heat always flows spontaneously from a hotter body to a colder body until both reach the same temperature.

Temperature does not depend on the amount of matter present, whereas the amount of heat transferred does. A bathtub of warm water and a cup of water at the same temperature can transfer very different amounts of heat.

To study heat flow, chemists divide the universe into two parts. The **system** is the specific portion of matter being studied, such as the chemicals reacting in a flask. The **surroundings** are everything outside the system that can exchange energy with it.

An **endothermic process** absorbs heat from the surroundings, so the surroundings become colder. Melting ice is endothermic. An **exothermic process** releases heat to the surroundings, so the surroundings become warmer. Burning wood is exothermic.

When a hot metal block is placed in cool water, heat flows from the block into the water until the temperature of the block equals the temperature of the water.

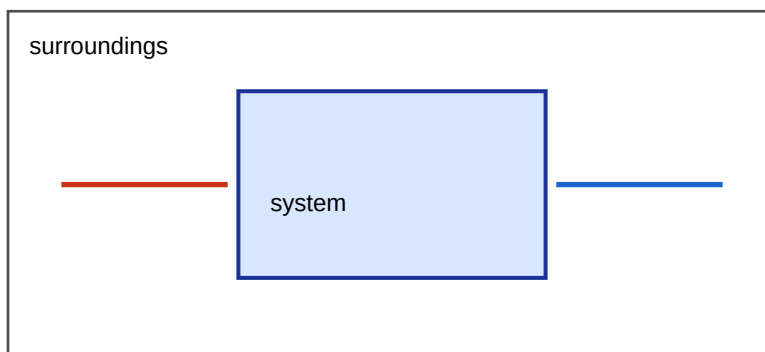


Figure 2.1 In an endothermic process, heat flows into the system from the surroundings. In an exothermic process, heat flows out of the system.

2.2 Specific Heat and Calorimetry

Learning Objectives

By the end of this section, you will be able to:

- Define heat capacity and specific heat capacity
- Calculate heat transferred using $q = m \times c \times \Delta T$
- Explain how a coffee-cup calorimeter measures heat

The **heat capacity** (C) of an object is the amount of heat needed to raise its temperature by 1 °C. Heat capacity depends on both the kind of substance and its amount.

The **specific heat capacity** (c) of a substance is the amount of heat needed to raise the temperature of 1 gram of the substance by 1 °C. Specific heat is an intensive property: it depends only on the kind of substance, not on the amount.

The heat absorbed or released by a sample that changes temperature without changing phase is given by

$$q = m \times c \times \Delta T \quad (2.1)$$

where q is the heat in joules, m is the mass in grams, c is the specific heat in J/(g·°C), and ΔT is the change in temperature (final temperature minus initial temperature). When q is positive the sample absorbs heat; when q is negative the sample releases heat.

Table 2.1 Specific Heats of Common Substances at 25 °C

Substance	c (J/(g·°C))
water (liquid)	4.184
aluminum	0.897
iron	0.449
copper	0.385

Example 2.1 Heat Absorbed by Water

How much heat is absorbed when 250. g of water is heated from 20.0 °C to 35.0 °C?

Solution The temperature change is $\Delta T = 35.0\text{ °C} - 20.0\text{ °C} = 15.0\text{ °C}$. Then $q = m \times c \times \Delta T = (250.\text{ g})(4.184\text{ J/(g·°C)})(15.0\text{ °C}) = 15,690\text{ J} = 15.7\text{ kJ}$. Because q is positive, the water absorbed heat.

A **calorimeter** is a device used to measure the heat that flows during a chemical or physical process. In a simple coffee-cup calorimeter, a reaction takes place in water inside an insulated cup, and the temperature change of the water is used to calculate the heat released or absorbed by the reaction.

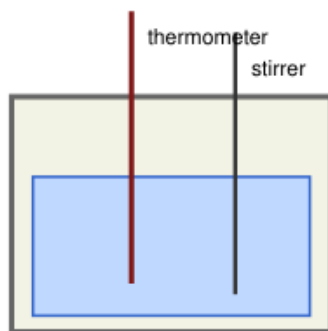


Figure 2.2 A coffee-cup calorimeter. Nested foam cups limit heat exchange with the surroundings.

Because the calorimeter is insulated, we assume that the heat released by the reaction equals the heat absorbed by the water: $q_{\text{reaction}} = -q_{\text{water}}$.

2.3 Enthalpy

Learning Objectives

By the end of this section, you will be able to:

- Define enthalpy and enthalpy change
- Relate the sign of ΔH to endothermic and exothermic processes

Most chemical reactions in the laboratory take place in open containers at constant atmospheric pressure. **Enthalpy** (H) is a property of a system equal to its internal energy plus the product of its pressure and volume:

$$H = U + PV \quad (2.2)$$

Chemists rarely measure enthalpy itself; instead they measure changes in enthalpy. The **enthalpy change** (ΔH) for a process carried out at constant pressure equals the heat absorbed or released by the system:

$$\Delta H = q_p \quad (2.3)$$

For an endothermic reaction, the system absorbs heat, so ΔH is positive. For an exothermic reaction, the system releases heat, so ΔH is negative. For example, the combustion of methane releases 890 kJ of heat per mole of methane burned, so $\Delta H = -890$ kJ.

Unlike heat, which depends on the path taken, enthalpy is a state function: ΔH depends only on the initial and final states of the system.

Example 2.2 Classifying a Reaction

The reaction $\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2 \text{NO}(\text{g})$ has $\Delta H = +180.5$ kJ. Is it endothermic or exothermic?

Solution ΔH is positive, so the reaction absorbs heat and is endothermic.

Key Terms

calorimeter device used to measure the heat that flows during a chemical or physical process

endothermic process process that absorbs heat from the surroundings

enthalpy (H) property of a system equal to its internal energy plus the product of its pressure and volume

enthalpy change (ΔH) heat absorbed or released by a system at constant pressure

exothermic process process that releases heat to the surroundings

heat (q) transfer of thermal energy between two bodies at different temperatures

heat capacity (C) amount of heat needed to raise the temperature of an object by 1 °C

specific heat capacity (c) amount of heat needed to raise the temperature of 1 gram of a substance by 1 °C

surroundings everything outside the system that can exchange energy with it

system specific portion of matter being studied

temperature measure of the average kinetic energy of the particles in a sample

Key Equations

$$q = m \times c \times \Delta T$$

$$H = U + PV$$

$$\Delta H = q_p$$

Summary

2.1 Heat and Temperature

Heat is the transfer of thermal energy between bodies at different temperatures, and temperature measures the average kinetic energy of particles. Endothermic processes absorb heat; exothermic processes release heat.

2.2 Specific Heat and Calorimetry

The specific heat capacity of a substance is the heat needed to raise the temperature of 1 g by 1 °C. The heat for a temperature change is $q = m \times c \times \Delta T$. Calorimeters measure heat flow.

2.3 Enthalpy

Enthalpy is $H = U + PV$. At constant pressure, the enthalpy change equals the heat absorbed or released. ΔH is positive for endothermic processes and negative for exothermic processes.

Review Questions

2.1 Heat and Temperature

5. Is the freezing of water endothermic or exothermic? Explain.
6. Identify the system and the surroundings when a reaction occurs in a beaker of water.

2.2 Specific Heat and Calorimetry

7. How much heat is needed to raise the temperature of 50.0 g of aluminum from 25.0 °C to 75.0 °C?
8. A 100.0-g sample of copper absorbs 385 J of heat. By how much does its temperature rise?

2.3 Enthalpy

9. The combustion of methane has $\Delta H = -890 \text{ kJ}$. Is heat absorbed or released?

Answer Key

Chapter 1

1. (b)
2. 9.00 J
3. 2092 J (2.09 kJ)
4. 239 cal

Chapter 2

5. Exothermic; liquid water releases heat as it freezes.
7. 2.24×10^3 J (2.24 kJ)
8. 10.0 °C
9. Released (exothermic).

Appendix A: SI Prefixes

kilo (k) = 10^3 ; mega (M) = 10^6 ; milli (m) = 10^{-3} ; micro (μ) = 10^{-6}